Mental Math Base Load — Cheatsheet

Goal: instant recall. Drill these daily for 5 minutes before method practice.

1. Multiplication Tables 1–15

×	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
3	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45
4	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60
5	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
6	6	12	18	24	30	36	42	48	54	60	66	72	78	84	90
7	7	14	21	28	35	42	49	56	63	70	77	84	91	98	105
8	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120
9	9	18	27	36	45	54	63	72	81	90	99	108	117	126	135
10	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
11	11	22	33	44	55	66	77	88	99	110	121	132	143	154	165
12	12	24	36	48	60	72	84	96	108	120	132	144	156	168	180
13	13	26	39	52	65	78	91	104	117	130	143	156	169	182	195
14	14	28	42	56	70	84	98	112	126	140	154	168	182	196	210
15	15	30	45	60	75	90	105	120	135	150	165	180	195	210	225

Target: answer any fact from 1 × 1 to 15 × 15 in under 1 second.

2. Complements

To 10	To 100	To 1000
1 → 9, 2 → 8, 3 → 7, 4 → 6, 5 → 5	23 → 77, 38 → 62, 47 → 53, 64 → 36, 81 → 19	238 → 762, 476 → 524, 613 → 387, 829 → 171

Rule: subtract from the next round number, then adjust. Example: 1000 – 476 = 524.

3. Doubles and Halves

Skill	Examples	Why it matters
Double	18 → 36; 36 → 72; 125 → 250	Multiply by 4, 8, 20, 40
Halve	86 → 43; 250 → 125; 750 → 375	Multiply/divide by 2, 4, 5, 50
×5	Multiply by 10, then halve: 48 × 5 = 480 ÷ 2 = 240	Fast 5% and ×5
×25	Divide by 4: 320 × 25 = 320 ÷ 4 × 100 = 8000	Fast 25%

4. Squares 1–25

n	1	2	3	4	5	6	7	8	9	10	11	12	13
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169
n	14	15	16	17	18	19	20	21	22	23	24	25	
n^2	196	225	256	289	324	361	400	441	484	529	576	625	

Also know: numbers ending in 5: $35^2 = 3 \times 4$ then append 25 = 1225.

5. Fractions, Decimals, Percentages

Fraction	Decimal	Percent	Fraction	Decimal	Percent
1/2	0.5	50%	1/8	0.125	12.5%
1/3	0.333	33.3%	3/8	0.375	37.5%
2/3	0.667	66.7%	5/8	0.625	62.5%
1/4	0.25	25%	7/8	0.875	87.5%
3/4	0.75	75%	1/10	0.1	10%
1/5	0.2	20%	1/20	0.05	5%
2/5	0.4	40%	1/16	0.0625	6.25%

Fraction	Decimal	Percent	Fraction	Decimal	Percent
3/5	0.6	60%	1/6	0.167	16.7%
4/5	0.8	80%	1/9	0.111	11.1%

6. Percentage Anchors

Percent	Method	Example
10%	Divide by 10	10% of 240 = 24
5%	Half of 10%	5% of 240 = 12
20%	Double 10%	20% of 240 = 48
25%	Divide by 4	25% of 240 = 60
50%	Halve	50% of 240 = 120
75%	Half + quarter	75% of 240 = 120 + 60 = 180
Any %	Break into 10% + 5% + 1% parts	17% of 240 = 24 + 12 + 4.8 = 40.8

7. Core Methods

Operation	Method	Example
Addition	Left to right; add tens first, then units	47 + 38: 40+30=70; 7+8=15; 85
Subtraction	Left to right; use complements	83 – 47: 83–40=43; 43–7=36
Multiplication	Split numbers; multiply parts; add	23 × 7 = 140 + 21 = 161
2-digit × 2-digit	Split one number: tens + units	47 × 23 = 47×20 + 47×3 = 940+141=1081
Near 100	Use base method	98 × 97 = 9506
Division	Think reverse multiplication; split divisor	456 ÷ 12 = 456 ÷ 4 ÷ 3 = 38
Decimal ×	Ignore point, multiply, restore places	0.3 × 0.4 = 0.12
Decimal ÷	Move decimal to make divisor whole	7.2 ÷ 0.6 = 72 ÷ 6 = 12

8. Daily 5-Minute Base Drill

Minute	Drill
1	Multiplication tables 1–15, random order
2	Complements to 100 and 1000
3	Squares 1–25
4	Fraction ↔ decimal ↔ percent conversions
5	Percent anchors: 5%, 10%, 20%, 25%, 50%, 75%

Accuracy rule: stay above 90% correct. If not, slow down and repeat the same set.